

Structured Masking Strategies for nano4M on MultiCLEVR

Comparing random, span, block, and mixed masking for multimodal pretraining

COM-304 Foundation Models · EPFL · Project Presentation

Model

nano4M / FourM

Dataset

MultiCLEVR

Key idea

Same architecture, different
masking

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Presentation Outline

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Qualitative analysis and discussion

Project Extension: From Random to Structured Masking

.Baseline nano4M uses random masking across all modalities -

Text has sequential structure → contiguous span masking -
.may be more meaningful

Image tokens have spatial structure → rectangular block -
.masking may be more meaningful

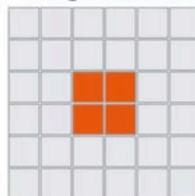
Research question: does respecting natural structure improve
?learning and generation

Text span



Modality	Structure	Natural masking idea
scene_desc	Sequential text	Mask contiguous spans
RGB / depth / normal	16×16 spatial grid	Mask rectangular blocks
All modalities	Multimodal alignment	Mix random + structured

Image block (16×16 grid, illustrative)



Training Setups and Compared Models

Shared experimental setup

Model **FourM d6-6w512**

Encoder / decoder depth **6 / 6**

Model dimension **512**

Dataset **MultiCLEVR**

Modalities **RGB · depth · normal · scene_desc**

Total tokens **5 000 M**

Batch size **256 per GPU**

Learning rate **6×10^{-4}**

Precision **bf16**

Model	Text masking	Image masking	Description
Baseline	random	random	original nano4M
V1	span	random	structured text only
V2	random	block	structured image only
V3	span	block	structured text + image
V4	span / random	block / random	mixed strategy

*Architecture, data, and main hyperparameters are kept constant.
The masking strategy is the only variable.*

Masking Strategies

Random masking

Selects target positions uniformly
- .at random
- .Used by the baseline



Simple, general, but ignores
- .modality structure

Span masking (text)

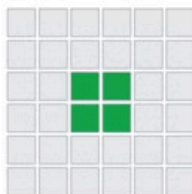
Masks contiguous text spans in
- .scene_desc
- .Geometric distribution, $p = 0.2$



Average span ≈ 5 tokens;
- .aligned with language

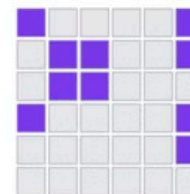
Block masking (image)

Masks rectangular regions in the
- .16x16 token grid
Applied to tok_rgb / depth /
- .normal
Aligned with spatial image
- .structure



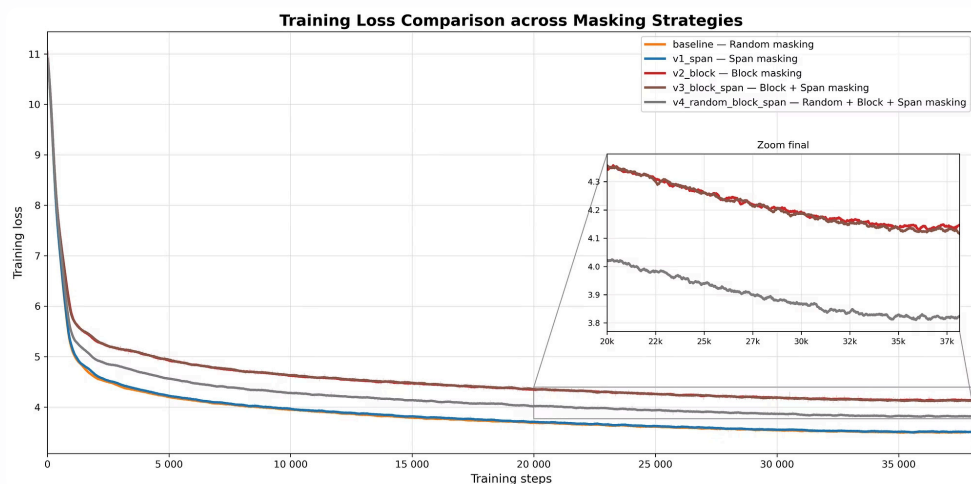
Mixed masking (V4)

Combines random and
- .structured masking
- .V4 uses a 50 / 50 mixture
- .Balances diversity and structure



Training Loss Comparison

Training loss across masking strategies, with a zoom on the final training phase



Key observations

- Training is stable for all five masking strategies.
- Baseline and V1 reach the lowest training loss.
- V4 remains close but higher; V2/V3 are harder objectives.
- Loss is not perfectly aligned with generation quality.

Takeaway: training loss is useful to check convergence, but the generation metrics decide which masking strategy is best.

Why We Needed a Custom Evaluation Pipeline

Task: image-to-text generation

Input modalities: tok_rgb@256, tok_depth@256,
tok_normal@256

Output modality: scene_desc

Original training setup provided checkpoints and loss curves.

Training loss alone was insufficient to compare masking strategies.

We needed generation-based evaluation on the same validation subset.

Problem

- Checkpoints only
- Training loss only
- No direct generated-text comparison
- No qualitative sample inspection

Need

- Generate predictions
- Score them automatically
- Compare all variants fairly
- Inspect outputs visually

❏ Training loss shows convergence, but generation-based evaluation is needed to compare actual model behavior.

EVALUATION

How the Evaluation Works



IMPLEMENTATION

What I Implemented

`generate_scene_desc_predictions.py`

Purpose: Generate scene_desc predictions from a checkpoint

- Loads a trained .pth checkpoint
- Runs inference on validation samples
- Saves reference/prediction pairs

`score_scene_desc_predictions.py`

Purpose: Compute automatic text-generation metrics

- BLEU-4, CIDEr, ROUGE-L, Exact Match
- Aggregates results over 200 validation samples

`compare_one_sample_all_models.py`

Purpose: Compare all masking variants on the same sample

- Baseline, V1, V2, V3, V4 side by side
- Exports a comparison report

`render_scene_desc_to_png.py`

Purpose: Create a visual rendering from a structured scene description

- Converts scene_desc into a CLEVR-like PNG
- Supports qualitative comparison of predictions

Image-to-Text Results and Conclusions

Model	BLEU-4	CIDEr	ROUGE-L	Exact Match
Baseline	69.59	37.07	84.31	4.00
V1	67.60	35.38	83.03	3.50
V2	68.70	34.40	83.67	4.00
V3	68.51	33.58	83.72	2.00
V4	69.64	37.55	84.07	4.00

Main Results

- V4 achieves the best overall trade-off
- Best CIDEr: V4 (37.55)
- Best BLEU-4: V4 (69.64)
- Best ROUGE-L: Baseline (84.31)
- Exact Match is tied between Baseline, V2, and V4

Interpretation

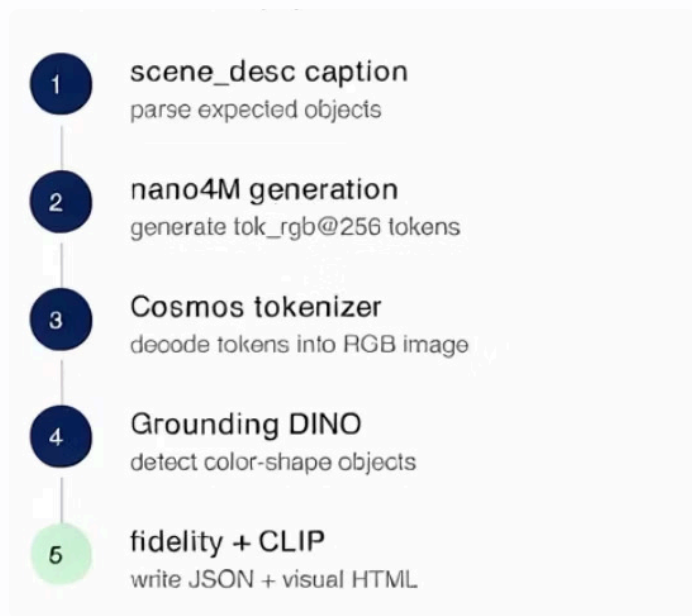
- V1 and V2 provide intermediate performance
- V3 is the weakest structured masking variant
- Fully structured masking may over-constrain generation
- Most qualitative differences are small coordinate shifts
- Shape, color, and material are usually predicted correctly

 Overall, mixed masking (V4) is the most robust strategy for image-to-text generation in this setting.

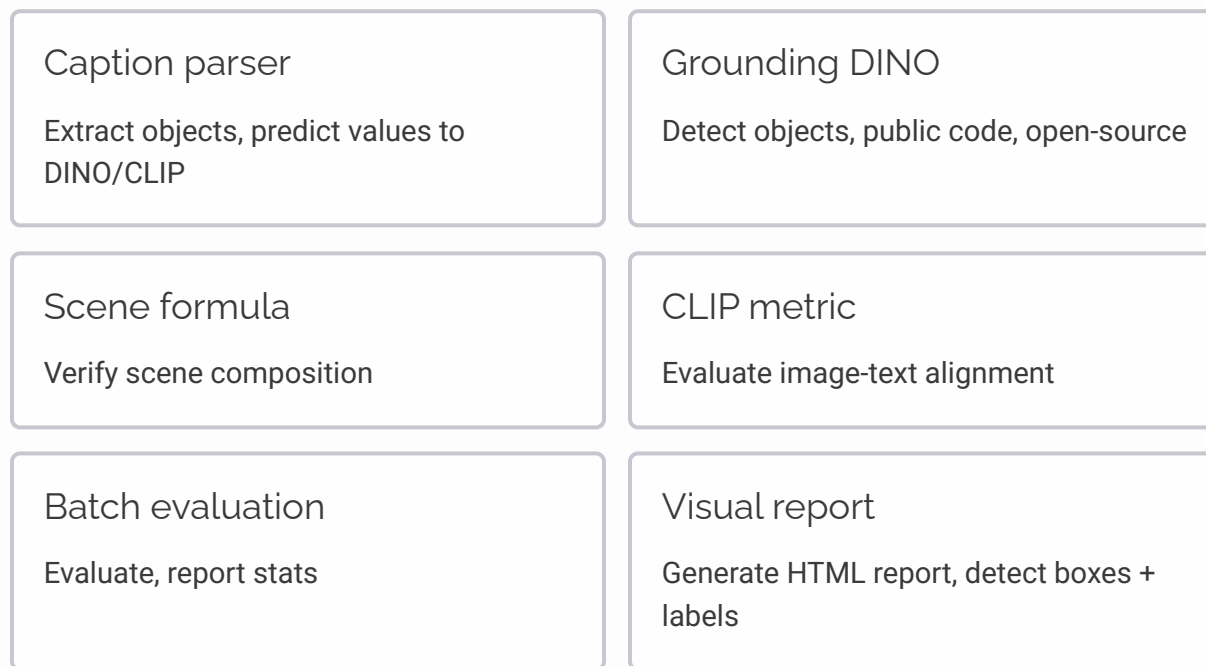
What I Built: CLEVR Scene Verifier

Goal: evaluate whether a generated image matches the requested scene composition.

End-to-end pipeline



Implemented pieces



❏ Beyond training loss: object-level scoring + visual report for manual inspection.

Text-to-Image Results: V1 is Best

CLEVR fidelity is the main metric. CLIP is reported as a complementary sanity check.

✓ Best model

V1

text-span / image-random

CLEVR fidelity

0.4735

CLIP

0.6301

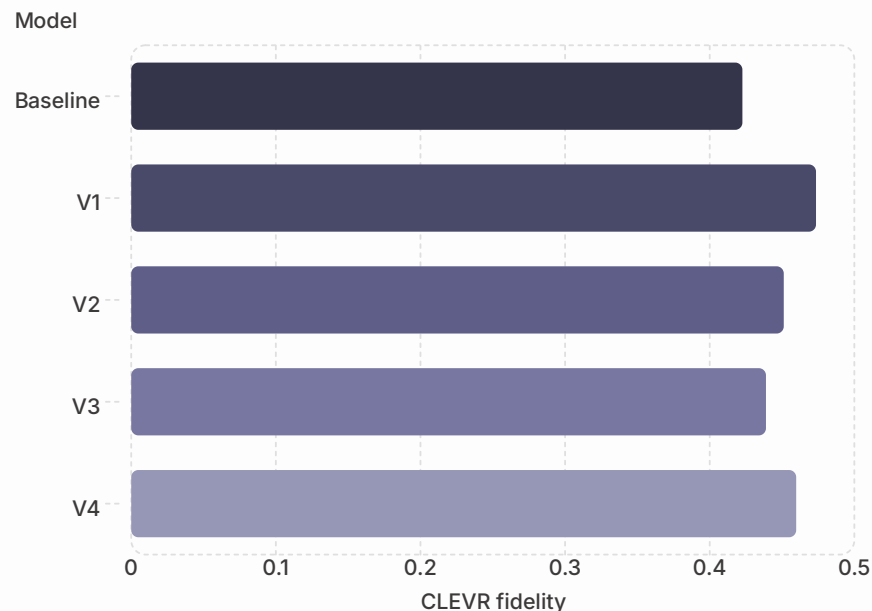
Gain over baseline

+0.0509 CLEVR

CIP range

0.6260-0.6301

CLEVR fidelity by model



Text-to-Image Generation Results

CLEVR fidelity & CLIP score on 200 validation samples

Model	CLEVR ↑	CLIP ↑	Std (CLEVR)
Baseline	0.4226	0.6287	0.298
V1 text-span / img-random	0.4735	0.6301	0.287
V2 text-random / img-block	0.4440	0.6275	0.304
V3 text-span / img-block	0.4340	0.6271	0.299
V4 mixed	0.4266	0.6260	0.311



V1 wins both metrics

Text-span + image-random

CLEVR: 0.4735

+0.0509 over baseline

CLIP: 0.6301

Best across all variants

*All four variants improve over baseline
on CLEVR fidelity.*

Ranking by CLEVR fidelity: V1 > V2 > V3 > V4 > Baseline

Why V1?

Text-span masking teaches contiguous-token recovery — aligned with image-token generation.

Image-random keeps visual context unrestricted, so the encoder sees diverse spatial cues during training.

Interpretation and Discussion

What the results suggest about structured masking in multimodal pretraining

What worked

- V4 is best for image-to-text: highest CIDEr and BLEU-4.
- V1 is best for text-to-image: highest CLEVR fidelity and CLIP.
- Structured masking helps when aligned with the target direction.

What did not work

- V3 is the weakest structured variant for image-to-text.
- Block masking alone does not surpass the random baseline in text metrics.
- Fully structured masking can make reconstruction too constrained.

Main lesson

- There is no universal best masking strategy.
- Randomness remains useful for diversity and robustness.
- The objective should match the generation direction and modality structure.

i Synthesis: structure is useful, but the best results come from balancing modality-aware masking with enough randomness to keep the task diverse.

CLOSING SLIDE

MULTICLEVR

STRUCTURED MASKING

Thank You

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 Project

Structured Masking Strategies for nano4M on MultiCLEVR

 Course

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We appreciate your time and attention.